

Codegen Pipeline (zero-drift contract)

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Master technical specification — Volume 6 (Deployment, Tooling & Operations), nano-detail document. Scope: the **schema-first code generation pipeline** — buf → Go/Dart/Rust from the canonical `.proto`, OpenAPI → Dart/TS/Go from the canonical REST spec, the zero-drift guarantee (generated code is regenerated + diff-gated), the codegen gates wired into the local pre-build (§11.4.156 — NO server CI), and concrete `buf.gen.yaml` / `Makefile` / drift-check sketches. This deepens [05-overview §4] and operationalizes the codegen seams declared in [protobuf-spec §6] and [svc-api §12]. It is a SPEC: configs and skeletons are illustrative of the contract, not the shipping implementation (2–3 refinement passes follow).

Evidence cited inline by id: [05_OV §N] = 05-repo-layout-tooling-and-helix-ecosystem.md; [PROTO §N] = v03-control-plane/protobuf-spec.md; [API §N] = v03-control-plane/svc-api.md; [04_ARCH §4.2] = schema-generated clients, no drift; [04_P1 §8] = codegen (buf + OpenAPI); [research-go_cp] = the Go control-plane research digest. Unproven facts are marked **UNVERIFIED** per §11.4.6 — never fabricated. The proto package is `helix.coordinator.v1`, unified across the spec set [PROTO §0.2].

0. What this document owns (and does not)

This document owns the **codegen mechanics**: the two generator families (buf for protobuf, OpenAPI generators for REST), their config files, the offline-reproducibility decision, the `make gen / verify-gen` contract, the breaking-change gates (buf `breaking` + `oasdiff`), the track-vs-gitignore decision for generated code, and how the gates wire into the local pre-build sweep (§11.4.110/.156).

It does **not** own: the `.proto` field-by-field semantics (canonical in [PROTO §1]); the REST route table / request schemas (canonical in [API §3/§4]); the FFI bridge surface (Volume 4 `ffi-surface.md`); the repo layout that places `helix-proto/` (that is `repo-layout-and-decoupling.md`). Those are referenced, never redefined.

1. Why schema-first: the anti-drift thesis

The single most important anti-drift mechanism in HelixVPN is that **all** Go/Dart/Rust/TS clients are *generated* from one canonical schema; hand-written clients are **forbidden** so the four codebases cannot drift apart [04_ARCH §4.2, 04_P1 §8, PROTO §6]. The contract has two canonical sources, each generating its language family:

flowchart LR

```
P["proto/helix/coordinator/v1/*.proto<br/>(agent plane – canonical [PROTO §1])"]
  O["openapi/helix-rest.v1.yaml<br/>(app plane – canonical [API §3/§4])"]
    P -->|buf generate| GO["Go server stubs (connect-go)"]
    P -->|buf generate| DART["Dart agent client"]
    P -->|buf generate| RUST["Rust edge/connector client (prost+tonic)"]
    P -->|buf lint + buf breaking| G1["GATE: wire-incompat → pre-build FAIL"]
    O -->|oapi-codegen| GOAPI["Go Gin server stubs"]
    O -->|openapi-generator / swagger_dart| DARTAPP["Dart app REST client"]
    O -->|openapi-typescript| TS["TS Console client"]
    O -->|oasdiff| G2["GATE: breaking REST change → FAIL"]
    G1 --> PB["pre-build §11.4.110/.156"]
    G2 --> PB
    classDef s fill:#ff9d0022,stroke:#ff9d00;
    class P,0 s;
```

The guarantee is **mechanical, not aspirational**: regenerate, then `git diff --exit-code` the generated tree — a stale committed (or stale gitignored-but-required) client FAILs the gate (§11.4.108 runtime-signature). This document specifies exactly that machinery.

2. buf for protobuf (agent plane)

The single Coordinator service `.proto` lives in `helix-proto` (`proto/helix/coordinator/v1/coordinator.proto`) and generates **three** language stubs over **Connect** (gRPC + gRPC-Web + Connect on one listener) so native agents and a future WASM Console use the same generated stubs [PROTO §2/§6, 04_P1 §3].

2.1 buf.yaml (module + lint + breaking)

```
# helix-proto/buf.yaml
version: v2
modules:
  - path: proto
lint:
  use:
    - STANDARD
```

```

except:
  - PACKAGE_VERSION_SUFFIX    # we use helix.coordinator.v1
    (suffix present) – documented exception [PROTO §6.1]
breaking:
  use:
    - FILE                    # field-number/type changes within
      v1 are breaking → gate (§5) [PROTO §6.1]

```

2.2 buf.gen.yaml (three language outputs)

Connect is the binding so ONE service serves native agents over HTTP/2 AND browser clients over gRPC-Web [04_P1 §3, PROTO §6.2]:

```

# helix-proto/buf.gen.yaml – remote-plugin convenience form
version: v2
plugins:
  # ---- Go (server) ----
  - remote: buf.build/protocolbuffers/go
    out: gen
    opt: paths=source_relative
  - remote: buf.build/connectrpc/go                # Connect Go
    handlers/clients
    out: gen
    opt: paths=source_relative
  - remote: buf.build/bufbuild/protovalidate-go    # buf.validate
    field constraints [API §5.3]
    out: gen
    opt: paths=source_relative
  # ---- Dart (client core boundary) ----
  - remote: buf.build/protocolbuffers/dart
    out: gen/dart
  - remote: buf.build/connectrpc/dart              # connect-dart
    client (UNVERIFIED plugin coord, §2.4)
    out: gen/dart
  # ---- Rust (edge / connector) ----
  - remote: buf.build/community/neoeinstein-prost  # prost
    messages
    out: gen/rust/src
  - remote: buf.build/community/neoeinstein-tonic  # tonic
    service stubs
    out: gen/rust/src

```

2.3 protovalidate constraints travel with the schema

Field constraints (wg_pubkey exactly 32 bytes, enroll_token min length, kind defined-only) live in the .proto via buf.validate and are enforced by a validateInterceptor server-side [API §5.3]; they generate into all three languages, so the same constraint cannot drift between client and server. Example, overlaid on the canonical messages [PROTO §1]:

```

import "buf/validate/validate.proto";

message EnrollRequest {

```

```

string    enroll_token = 1
          [(buf.validate.field).string.min_len = 16];
bytes     wg_pubkey     = 2 [(buf.validate.field).bytes.len =
32];    // exactly 32B Curve25519 (C6)
string    os             = 3
          [(buf.validate.field).string.max_len = 32];
string    name           = 4
          [(buf.validate.field).string.max_len = 128];
DeviceKind kind          = 5
          [(buf.validate.field).enum.defined_only = true];
}

```

2.4 UNVERIFIED plugin coordinates

The remote plugin coordinates/versions above are the **conventional** buf plugin names; the **pinned** versions are chosen at implementation time and recorded in `buf.gen.yaml + buf.lock` then — they are not asserted here as fixed [PROTO §6.2]. The `connectrpc/dart` plugin coordinate in particular is **UNVERIFIED** (the Dart Connect ecosystem moved during the spec window); the fallback is plain `protocolbuffers/dart` messages + a thin hand-checked-but-generated transport shim, which keeps the message types generated even if the service stub is not.

3. OpenAPI for REST (app plane)

The app-facing REST surface is the canonical `openapi/helix-rest.v1.yaml` [API §4/§12]. Three generated consumers, zero hand-written app clients:

Target	Generator	Consumes
Go server stubs (Gin)	<code>oapi-codegen</code>	<code>helix-go/internal/api</code> handler signatures + request/response models
Dart app REST client	<code>openapi-generator</code> (or <code>swagger_dart_code_generator</code>)	the three Flutter apps (Access/Connector/Console)
TS Console client	<code>openapi-typescript</code>	the Console Web build

3.1 Spec-first vs code-first (decision)

Decision D-OPENAPI-AUTHORING. The OpenAPI spec is authored **spec-first** (hand-authored contract, handlers validated against it), NOT generated from handler annotations (code-first). Rationale [API §12.2, `research-go_cp` §1]: Gin is the thin edge; authoring the contract first lets the TS/Dart app clients generate from a stable surface, and a request-validation middleware (`kin-openapi`) then asserts every request/response conforms — a drift between handler and spec FAILs a test. **UNVERIFIED** until the

validation middleware is wired whether the MVP enforces response-conformance as well as request-conformance; the lean path enforces request-conformance first.

oapi-codegen config (server stubs) is referenced from the Makefile (§5):

```
# helix-go/internal/api/oapi.cfg.yaml (illustrative)
package: api
generate:
  gin-server: true
  models: true
  embedded-spec: true      # embed the spec so /v1 self-
                           # describes; never re-authored at runtime
output: internal/api/oapi_gen.go
```

3.2 The REST breaking-change gate

oasdiff compares the working helix-rest.v1.yaml against the last released git ref; a breaking REST change (removed route, narrowed type, required-field addition) FAILs the gate [API §12]. This is the REST analogue of buf breaking (§5).

4. Track or gitignore the generated code? (the §11.4.77 decision)

Decision D-CODEGEN-TRACK. Generated clients are **gitignored**, with make gen as the §11.4.77 regeneration mechanism (NOT committed). Rationale: the schema is the single source of truth (§11.4.93 spirit applied to code); a committed generated tree is a second copy that can rot and doubles review noise; make gen from a clean clone reproduces it deterministically. The gitignore entry carries a .gitignore-meta/<slug>.yaml declaring the mechanism (§11.4.77), and scripts/setup.sh runs make gen-tools && make gen post-clone so a fresh checkout builds.

This **refines** the overview, which kept generated code gitignored (helix-proto/gen/ is gitignored [05_OV §2]) — this document makes that the binding decision and pins the §11.4.77 mechanism. The trade-off: a fresh clone cannot go build until make gen runs once; the post-clone bootstrap (scripts/setup.sh) closes that gap, and the §11.4.77 meta-file makes the regeneration mechanism auditable.

UNVERIFIED: [svc-api §12.1] notes "generated Go/Dart/Rust stubs are committed (regeneratable) so the three codebases cannot drift" — i.e. the *control-plane* doc leaned track-committed. This is a **visible reconciliation** (§11.4.35): the binding Volume-6 decision is **gitignore + regenerate** (D-CODEGEN-TRACK), with the drift gate (§6) providing the no-drift guarantee that committing was meant to provide. Either choice satisfies §11.4.108 (the runtime signature is "regenerate → diff is empty"); gitignore + the §11.4.77 mechanism is chosen for the smaller review surface. The reconciliation is recorded so the two docs do not silently disagree.

5. The breaking-change gates (frozen wire + frozen REST)

Two gates make the contract frozen-by-machine, run in the local pre-build sweep (§11.4.156 — NO server CI; enforcement is the local pre-build + git hooks, §11.4.75):

```
# agent plane (buf)
buf lint                                # STANDARD rules
buf breaking --against '.git#branch=main' # field renumber /
      type-change / reused-number → FAIL
buf generate --template buf.gen.local.yaml # emits gen/ (Go),
      gen/dart, gen/rust

# app plane (OpenAPI)
oasdiff breaking helix-rest.v1.yaml@main helix-rest.v1.yaml #
      breaking REST change → FAIL
```

buf breaking is the mechanical enforcement that the [PROTO §1.1] field-number registry stays frozen: any commit renumbering a v1 field, changing a field type, or reusing a removed number FAILs the gate. Paired §1.1 mutation: a branch that renames Peer.allowed_ips field number 3 → 7 MUST make buf breaking FAIL; reverting MUST make it pass [PROTO §6.3].

6. The make gen / verify-gen contract (the drift gate)

6.1 Offline reproducibility (the §11.4.77 belt)

Decision D-CODEGEN-NET. buf remote plugins need network. Per §11.4.77 the regeneration mechanism MUST work from a clean clone. **Recommended:** vendor protoc-gen-* plugins as pinned local binaries under tools/ (Go via tools.go + go install, Rust via cargo install --locked) so make gen runs **offline** after a one-time make gen-tools; buf generate --template buf.gen.local.yaml points at the local binaries. The remote-plugin form (§2.2) stays as the convenience path [05_OV §4.1].

```
# Makefile – codegen targets (§11.4.156 NO CI)
gen-tools:                                ## one-time: install pinned local protoc-
      gen-* + oapi-codegen + openapi gens
go install github.com/bufbuild/buf/cmd/buf@<pin>
go install github.com/oapi-codegen/oapi-codegen/v2/cmd/oapi-
      codegen@<pin>
cargo install --locked protoc-gen-prost protoc-gen-tonic
# dart/ts generators per their pinned versions

gen: gen-proto gen-openapi gen-ffi      ## regenerate ALL
      clients from schema (run before build)
gen-proto: ; buf generate --template buf.gen.local.yaml
```

```

gen-openapi: ; oapi-codegen -config helix-go/internal/api/
              oapi.cfg.yaml openapi/helix-rest.v1.yaml \
              && openapi-generator generate -i openapi/helix-
rest.v1.yaml -g dart -o helix-ui/packages/
helix_api_client \
              && npx openapi-typescript openapi/helix-
rest.v1.yaml -o helix-ui/packages/app_console/src/gen/
api.ts
gen-ffi:      ; cd helix-core/crates/helix-ffi &&
              flutter_rust_bridge_codegen generate

```

<pin> placeholders are resolved to exact versions at implementation time and recorded in `buf.lock + go.mod + Cargo.lock` — they are not asserted as fixed here (**UNVERIFIED** exact versions, per §11.4.6).

6.2 verify-gen — the drift gate itself

```

verify-gen: gen
            ## §11.4.108 artifact check: regen MUST be a no-op
            @git diff --exit-code helix-proto/gen helix-ui/packages/
            helix_core_ffi helix-ui/packages/helix_api_client \
            || { echo "GENERATED CLIENTS STALE – run make gen and commit
            the schema change"; exit 1; }

```

`verify-gen` is the **drift gate** (the local equivalent of a CI check, run in the pre-commit hook per §11.4.75): regenerate, then assert the tree is byte-identical. Because generated code is gitignored (D-CODEGEN-TRACK), `verify-gen` runs against a freshly-generated tree and asserts the generation is **deterministic** (two runs identical) AND that the schema → code mapping has no manual edits creeping in via the FFI/oapi hand-config. This is the schema-first guarantee made mechanical [05_OV §4.3].

6.3 The full gate wiring into pre-build (§11.4.110)

flowchart TD

```

    EDIT[edit .proto OR openapi.yaml] --> PC[pre-commit hook
§11.4.75]
    PC --> L1[buf lint]
    PC --> L2[buf breaking vs main]
    PC --> L3[oasdiff breaking vs main]
    L1 --> GEN[make gen]
    L2 --> GEN
    L3 --> GEN
    GEN --> VG[verify-gen: git diff --exit-code]
    VG -->|clean| OK[commit allowed]
    VG -->|drift| FAIL[BLOCK commit – regen + re-stage]
    OK --> PBV[pre-build verification §11.4.110 READY-FOR-BUILD]

```

The pre-build READY-FOR-BUILD verdict (§11.4.110) refuses to start the main build unless `buf lint`, `buf breaking`, `oasdiff`, and `verify-gen` are all green — codegen drift is a build-blocker, not a warning.

7. Cross-language round-trip proof (no-stub-drift)

The strongest anti-bluff for "the stubs actually interoperate" is a **cross-language round-trip**: a generated Rust tonic client and the generated Go connect-go server complete one real WatchNetworkMap exchange [PROTO §7 T18]. This catches a class of drift that verify-gen cannot — a generation that is internally consistent in each language but semantically divergent across languages (e.g. a oneof mapped differently).

sequenceDiagram

participant RS as Rust tonic client (gen/rust)

participant G0 as Go connect-go server (gen/go)

RS->>G0: WatchNetworkMap(WatchRequest{device_id,
known_version=0})

G0-->>RS: MapUpdate{version=42, snapshot}

Note over RS,G0: assert decoded snapshot fields == sent server-side state
(prevents stub drift; captured transcript)

A Dart client variant proves the third language. Infra (Postgres+Redis) is booted on demand via vasic-digital/containers (§11.4.76), never ad-hoc docker run [PROTO §7].

8. Test points (§11.4.169 comprehensive test-type coverage)

Each row is implementation-ready; every PASS cites captured evidence (§11.4.5/.69/.107) — a green test that does not exercise the generated code is a §11.4 PASS-bluff.

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
C1	unit	generator determinism	make gen twice → byte-identical output (hash compare); evidence under docs/qa/<run-id>/
C2	unit	protovalidate constraints generated	the generated Go/Dart/Rust validators reject a 31/33-byte wg_pubkey (§2.3)
C3	contract / compat	buf breaking freeze (§5)	renumber Peer.allowed_ips 3 → 7 → buf breaking FAILs; revert → passes (paired §1.1)
C4	contract / compat	oasdiff REST freeze (§5)	remove a /v1 route from the spec → oasdiff FAILs
C5	contract / no-drift	verify-gen (§6.2)	hand-edit a generated stub → verify-gen git diff FAILs; regen restores
C6	integration	cross-lang round-trip (§7)	Rust tonic client [2] Go connect-go server complete WatchNetworkMap; captured transcript

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
C7	integration	OpenAPI  handler conformance (§3.1)	kin-openapi middleware: a request violating the spec → rejected; a handler diverging from spec → test FAIL
C8	offline-repro	make gen from clean clone, network off (§6.1)	container clone with no network → make gen-tools && make gen exits 0; transcript captured
C9	meta-test (§1.1)	every gate has a paired mutation	renumber field / hand-edit stub / remove route / inject volatile codegen → each makes its gate FAIL

9. Phase-2 forward seams (additive, no v1 break)

Codegen seams permit Phase-2 growth without a v1 break [PROTO §8, API §14]: new .proto fields (new field numbers, buf breaking stays green) regenerate into all three languages; new REST routes added spec-first regenerate the app clients; a WASM Console gains the same generated Connect stubs (gRPC-Web already supported [API §5.1]); the PQ pre-shared-key field is a buf-additive field on EnrollResponse/Peer. Each is additive because v1 drew the field-number registry [PROTO §1.1] with room and froze it under buf breaking (§5).

Sources verified

- 05-repo-layout-tooling-and-helix-ecosystem.md §4 (schema-first codegen — buf + OpenAPI, the make gen/verify-gen contract, D-CODEGEN-NET offline reproducibility) — [05_OV].
- v03-control-plane/protobuf-spec.md §1 (canonical .proto + field-number registry), §2 (Connect binding), §6 (buf pipeline + breaking gate), §7 (test points incl. T18 cross-lang), §8 (forward seams) — [PROTO].
- v03-control-plane/svc-api.md §4 (REST schemas), §5.3 (protovalidate), §12 (schema-first no-drift gates — buf + OpenAPI + oasdiff + git-diff runtime signature), §14 (forward seams) — [API].
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md §4.2 (schema-generated clients, no drift), HelixVPN-Phase1-MVP.md §8 (codegen), 02-control-plane.md / research-go_cp §1–§2 (Gin thin edge, connect-go, protovalidate, single-server) — [04_ARCH] [04_P1] [research-go_cp].
- Constitution anchors: §11.4.77 (regeneration mechanism for gitignored content), §11.4.108 (runtime-signature: regen-diff-empty), §11.4.110 (READY-FOR-BUILD pre-build verdict), §11.4.156 (no active CI — local pre-build enforcement), §11.4.76 (containers submodule for test infra), §11.4.35 (visible reconciliation — D-CODEGEN-TRACK vs the control-plane doc's track-committed note), §11.4.6 (no-guessing — UNVERIFIED plugin/version marks), §11.4.169 (test-type coverage), §1.1 (paired meta-test mutations).